

Data Analysis using SQL (Business Transactions)

1. Revenue by Gender : Compared total revenue generated by male vs. female customers.

	Gender	revenue
▶	Female	75191
	Male	157890

2. High-Spending Discount Users : Identified customers who used discounts but still spent above the average purchase amount.

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Customer ID	Purchase Amount (USD)			
▶	2	64			
	3	73			
	4	90			
	7	85			
	9	97			
	12	68			
	13	72			
	16	81			
	20	90			
	22	62			
	24	88			
	25	64			
customer 13 x					
Output					
Action Output					
#	Time	Action	Message		
1	16:22:09	SELECT 'Customer ID', 'Purchase Amount (USD)' FROM customer WHERE 'Discount Applied' = 'Yes' AN...	839 row(s) returned		

3. Top 5 Products by Rating: Found products with the highest average review ratings.

	Item Purchased	Average Review Rating
▶	Gloves	3.8627737226277383
	Sandals	3.8446540880503144
	Boots	3.818881118881119
	Hat	3.801307189542483
	Skirt	3.7853503184713366

4. Shipping Type Comparison: Compared average purchase amounts between Standard and Express shipping.

	Shipping Type	average_purchase
▶	Express	60.48
	Standard	58.46

5. Subscribers vs. Non-Subscribers: Compared average spend and total revenue across subscription status.

	Subscription Status	total_customers	avg_spend	total_revenue
▶	No	2847	59.87	170436
	Yes	1053	59.49	62645

6. Discount-Dependent Products: Identified 5 products with the highest percentage of discounted purchases.

	Item Purchased	discount_percentage
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. Customer Segmentation: Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	customers	revenue	avg_spend
▶	Loyal Customer	3476	207389	59.66
	Returning Customer	341	20839	61.11
	New Customer	83	4853	58.47

8. Top 3 Products per Category: Listed the most purchased products within each category.

	Category	Item Purchased	purchase_count	product_rank
▶	Accessories	Jewelry	171	1
	Accessories	Sunglasses	161	2
	Accessories	Belt	161	2
	Clothing	Blouse	171	1
	Clothing	Pants	171	1
	Clothing	Shirt	169	3
	Footwear	Sandals	160	1
	Footwear	Shoes	150	2
	Footwear	Sneakers	145	3
	Outerwear	Jacket	163	1
	Outerwear	Coat	161	2

9. Repeat Buyers & Subscriptions: Checked whether customers with >5 purchases are more likely to subscribe.

	customer_type	Subscription Status	customer_count	percentage
▶	Non-Repeat Buyer	Yes	95	22.41
	Non-Repeat Buyer	No	329	77.59
	Repeat Buyer	Yes	958	27.56
	Repeat Buyer	No	2518	72.44

10. Revenue by Age Group: Calculated total revenue contribution of each age group.

	age_group	customers	avg_spend	revenue
▶	51+	1476	59.95	88480
	18-25	571	60.65	34630
	36-50	1111	59.07	65629
	26-35	742	59.76	44342

11. Customer Lifetime Value (CLV) Analysis: Estimated customer value using purchase amount and previous purchase behavior, then segmented customers into Low Value, Medium Value, High Value, and VIP groups. Identified that VIP customers contributed 51.52% of total estimated CLV, enabling targeted retention and marketing strategies.

	CLV_Segment	customers	total_clv	contribution_percentage
▶	VIP	975	3049885	51.52
	High Value	973	1638803	27.68
	Medium Value	977	919556	15.53
	Low Value	975	311476	5.26

12. Customer Churn Risk Analysis: Developed a rule-based churn risk model using purchase frequency, subscription status, and customer review ratings to identify customers with potential retention risks. Combined churn risk with CLV segmentation to prioritize valuable customers, identifying 263 VIP and 431 High Value customers at high churn risk for targeted retention strategies.

13. Product Performance Analysis: Categorized products based on sales performance and customer ratings into Hero Products, Hidden Gems, Problem Products, and Low Performers to identify growth opportunities and improvement areas.

	CLV_Segment	Churn_Category	customers	value_at_risk
▶	VIP	High Risk	263	801332
	High Value	High Risk	431	703928
	Medium Value	High Risk	604	560735
	Low Value	High Risk	824	260249

	Item Purchased	Purchase Amount (USD)	Review Rating
▶	Blouse	10410	3.6804733727810652
	Shirt	10332	3.6220238095238093
	Dress	10320	3.75
	Pants	10090	3.7208333333333337
	Sunglasses	9649	3.751572327044025
	Scarf	9561	3.705806451612903
	Shorts	9433	3.714193548387097
	Coat	9275	3.727672955974843

14. Customer Persona Analysis: Developed customer personas using CLV segmentation, churn risk, subscription status, and discount behavior. Identified VIP Loyalists, Deal Seekers, Regular Shoppers, and At-Risk Customers to support targeted retention and marketing strategies.

	Customer_Persona	customers	total_clv	clv_percentage
▶	Deal Seeker	1414	1735881	29.32
	Regular Shopper	729	1684405	28.45
	At-Risk Customer	1494	1674751	28.29
	VIP Loyalist	263	824683	13.93

BUSINESS RECOMMENDATIONS

- Launch retention campaigns for VIP high-risk customers.
- Investigate problem products with low ratings.
- Use personalized discounts for deal seekers.
- Promote hidden-gem products.
- Increase loyalty incentives for repeat buyers.